AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter Monitoring & Health Effects

Issue 2026-09-09 | Entry window 9 September 2026 | 26 records screened in scope

26 RECORDS IN SCOPE (71 EXCLUDED)

9 SUBTOPIC CLUSTERS REPRESENTED

10 EFFECT ESTIMATES EXTRACTED

7 SENSING & INSTRUMENTATION

4 METADATA ONLY (NO ABSTRACT)

Signal of the day

The $PM_{2.5}$ component that carries the health signal is the one the field's standard instrument reads wrong.

Two records that arrived on the same day, from literatures that do not cite each other. Massabò et al. (*Atmosphere*) ran an AE33 aethalometer against two independent offline filter techniques — MWAA and BLAnCA — at an urban coastal site in Genoa. The three agree on *shape* (R^2 **0.93–0.99** across UV, visible and near-IR) and disagree on *level*: AE33 reports absorption **12–22% high**, wavelength dependent, which points at the default multiple-scattering correction rather than at noise. Meanwhile Yu et al. (*JAHA*), in a time-stratified case-crossover on nine years of Hangzhou acute aortic dissections, found that an IQR rise in black carbon carried a **42% increase** in type-A dissection odds (OR 1.42, 95% CI 1.15–1.75), ahead of organic matter (1.35) and sulfate (1.36) — and that BC alone took **over half the weighted-quantile-sum joint weight**. So the component-resolved epidemiology is converging on BC exactly where the routine BC measurement carries a double-digit multiplicative bias that varies by site and aerosol population. A 15% error in the exposure is not a rounding detail when the effect being estimated is a 42% odds increase per IQR. The third record closes the loop: the RI-URBANS synthesis (*Atmos. Meas. Tech.*) reports, across a continent of low-cost-sensor, mobile and citizen-science pilots, that the binding constraint on usable urban maps was never sensor accuracy — it was *harmonisation* between methods. For distributed sensing the implication is concrete and unwelcome: a dense BC network calibrated against a site-specific aethalometer offset is not comparable to anyone else's dense BC network.

What changed today

- **Personal and mobile monitoring produced the two strongest exposure records of the day, and they disagree about what resolution buys you.** MobiliSense (*preprint*) put two ambulatory monitors, a GPS and an accelerometer on 199 Paris adults and took 2,504 spirometry tests: BC one hour before testing cut FEV_1 by **0.016 L** (95% CI -0.024 to -0.008) per 1 $\mu g/m^3$, and $PM_{2.5}$ at 15–60 minutes cut FEV_1/FVC by 0.5–0.7 percentage points — effects that a daily residential average cannot resolve at all. The Hong Kong GEMA study (*Environ Res*) went further, 1.7 million personal $PM_{2.5}$ measurements on 800 adults across eight activity-travel contexts, and found that context-specific associations *reversed sign* while the pooled estimate showed nothing sustained. Higher resolution is not automatically a stronger signal; it can be a decomposition of a null into opposing parts.
- **Three clean nulls, all of them worth more than a marginal positive.** A 24-cohort, 126-million-participant meta-analysis (*Environ Pollut*) found no sex modification of the $PM_{2.5}$–Alzheimer's association (RRR 0.963, 0.919–1.009) or of vascular dementia (1.003, 0.954–1.054), with GRADE moderate certainty, and states plainly that sex-specific policy is not supported. A matched case-control with a distributed-lag model found the green-space–ASD protective window is real in mid-to-late pregnancy and that $PM_{2.5}$ *neither mediated nor modified* it — closing the obvious confounding pathway rather than leaving it open. And the smoke-free-home cluster RCT missed its primary outcome outright.
- **Non-exhaust and non-filterable sources both surfaced, from opposite directions.** Tire-wear particles inhibited shoot and root growth and raised root lipid peroxidation in a roadside grass (*Curr Biol*); condensable PM — which leaves the stack as vapour and forms particles downstream, so filterable-PM controls miss it entirely — was removed at 29.6–65.3% by a string-of-beads liquid at under a tenth the energy of condensation (*J Hazard Mater*). Both are sources that mass-based optical monitors size or miss badly.
- **The abstract-deposit gap is now a structural property of a one-day window, not an incident.** Four Elsevier records (*Environ Int*, *Atmos Environ*, *Atmos Pollut Res*, *Eur J Intern Med*) had no abstract in Crossref per-DOI, Europe PMC or PubMed on the entry date. They ship as METADATA ONLY, asserting no finding. This is the second consecutive issue to hit it; see the provenance note.
- **The PubMed connector added net-new records for the first time.** Its health axis returned 16 PMIDs against the harvester's own E-utilities leg's 11, nine of them unseen, three of which ship — an analytical method for atmospheric alkylamines at 0.1–1.1 pg/m^3 in PM, the condensable-PM removal study, and a cardiorespiratory review. On 8 September the connector was pure confirmation. It is not redundant.

Corpus analytics

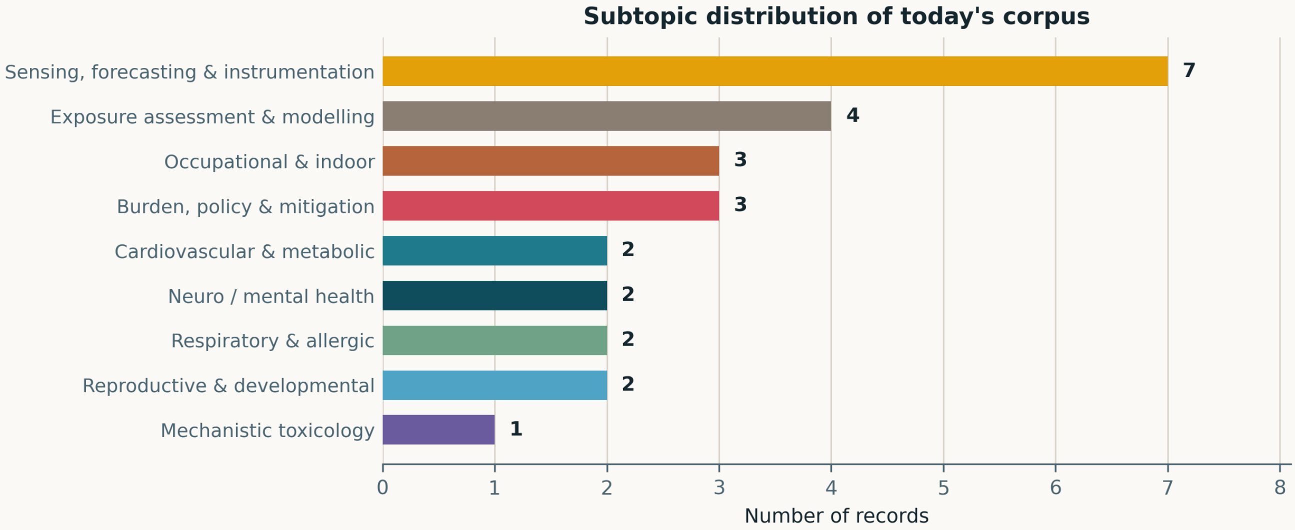
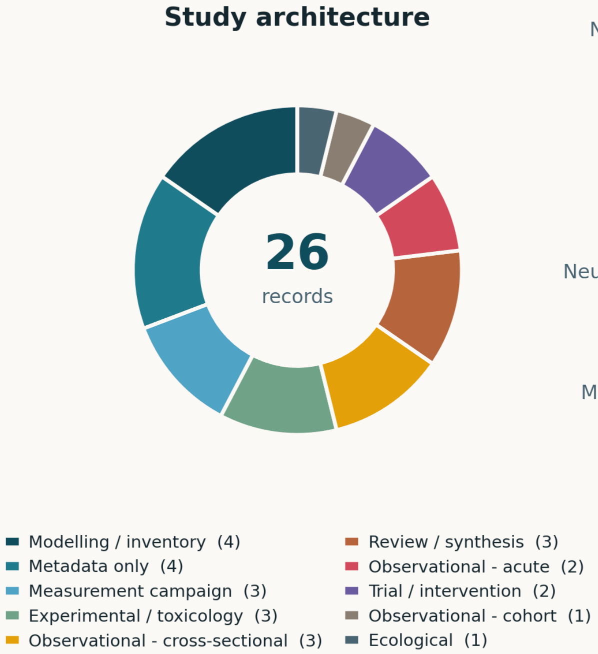


Fig. Subtopic assignment of the 26 in-scope records; single-label by dominant contribution. *Sensing, forecasting & instrumentation* is the largest cluster at 7, followed by *Exposure assessment & modelling* at 4 — together 42% of the issue, an instrumentation-weighted day. The four clinical clusters (cardiovascular/metabolic, neuro, respiratory, reproductive) carry two records each. *Other clinical endpoints* is the only empty cluster: **9 of 10 represented** ties 4 September for the broadest coverage of the past week, against 6 clusters on 8 September.



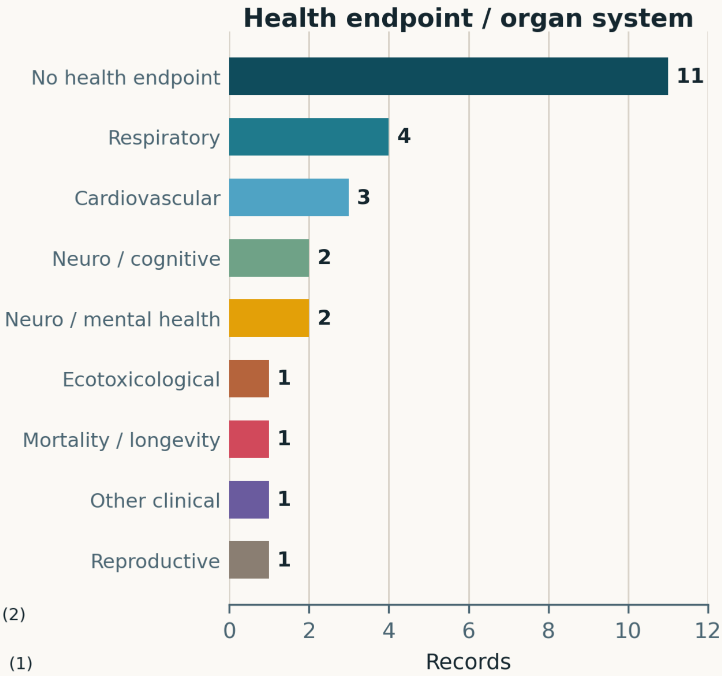


Fig. Left — study architecture. *Modelling / inventory* and *Metadata only* tie at 4 each; the three observational headings (cross-sectional 3, acute 2, cohort 1) sum to **6**, with one further *Ecological* record. Two trial records is unusual for a single day, and neither reports a positive primary result: one is a protocol, the other missed its endpoint. Right — health endpoint. **Eleven of 26 records carry no human health endpoint at all** — the monitoring, instrument, emissions and modelling records — against 4 respiratory, 3 cardiovascular and 2 each for mental health and neuro/cognitive. One record is ecotoxicological (plants, not people) and is labelled as such rather than folded into a human category.

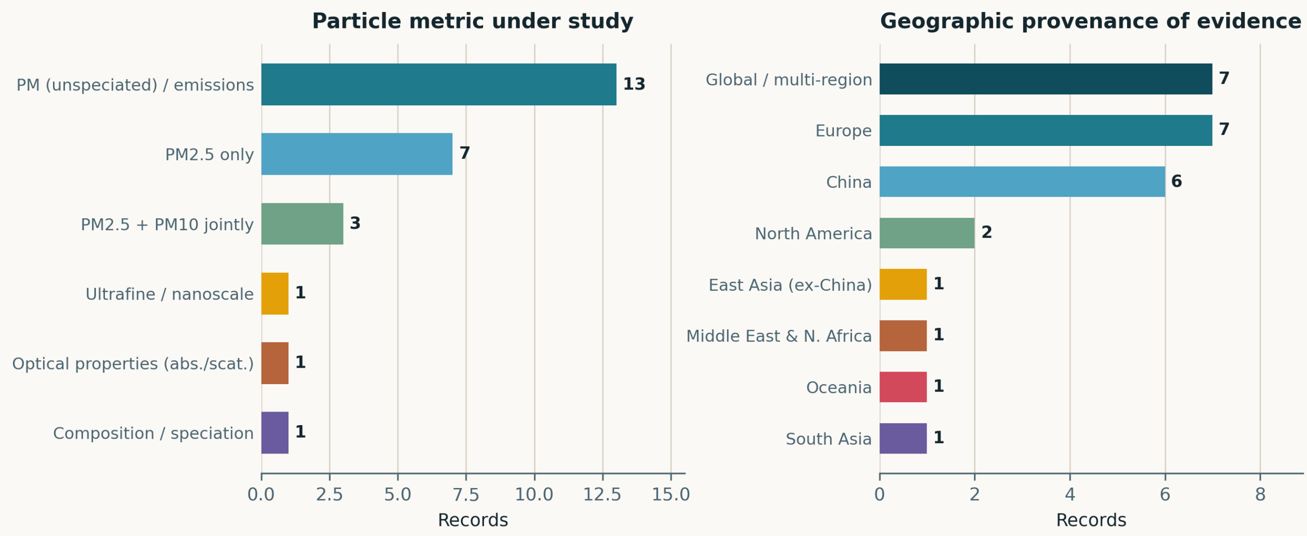


Fig. Left — particle metric. The *PM (unspeciated) / emissions* bar dominates at 13 because the day is instrumentation- and source-heavy; $PM_{2.5}$ alone is the exposure in 7 and joint $PM_{2.5}/PM_{10}$ in 3. Only **one** record resolves composition and one resolves optical properties — which is the gap the Signal of the day turns on. Right — geographic provenance. Europe and the *Global / multi-region* fallback tie at 7, but the fallback is not what it looks like: 3 are genuinely multi-region reviews and **4 are records whose study setting could not be established** because they have no abstract. China contributes 6. South Asia, Oceania, MENA and East Asia (ex-China) contribute one apiece; sub-Saharan Africa contributes none, against 3 on 8 September.

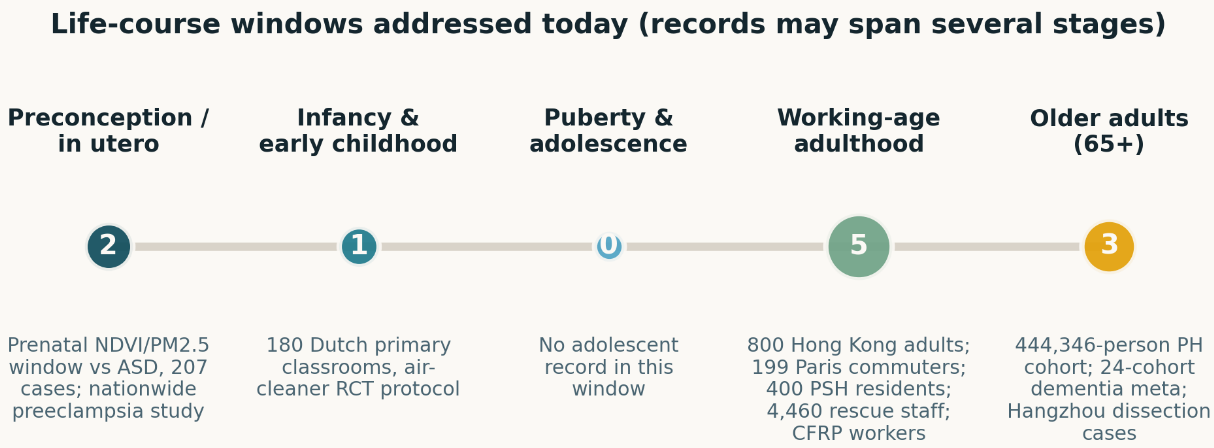


Fig. Life-course windows addressed; counts are non-exclusive and only the 11 records with an identifiable human window are placed. Working-age adulthood dominates at 5 — the Hong Kong and Paris mobility cohorts, the permanent-supportive-housing trial, the rescue personnel survey and the CFRP occupational review. Puberty and adolescence is empty, as it has been for most of this archive; the two in-utero records are the prenatal green-space window study and the preeclampsia record.

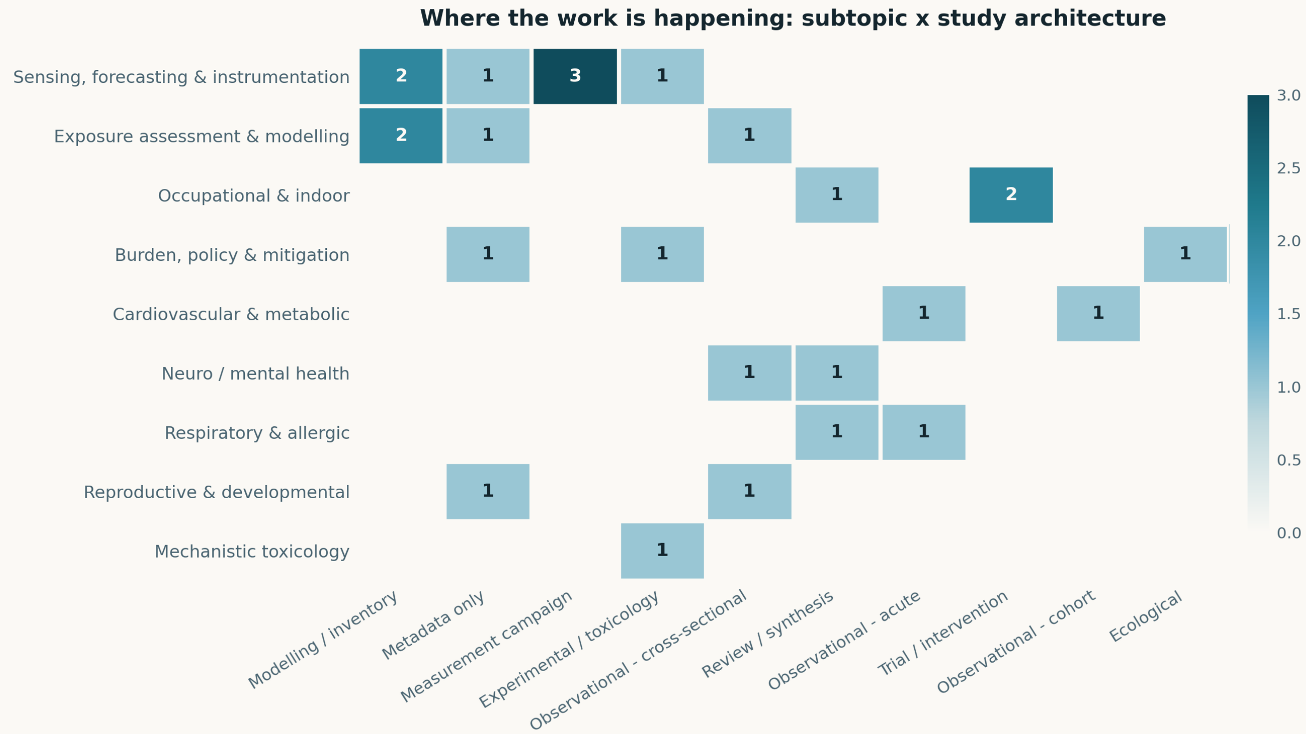


Fig. Subtopic × study-architecture cross-tabulation. The single densest cell is *Sensing* × *Measurement campaign* at 3 (the Genoa intercomparison, the Baltic ship-plume campaign and the RI-URBANS synthesis); three further cells tie at 2. The structural feature to note is the same one this archive keeps recording: **all seven *Sensing* records carry no health endpoint**, and three of the four *Exposure* records do not either — the sole exception in these two rows being the Hong Kong GEMA study. Only two records in the whole issue put a *measured personal* particle exposure and a human outcome in the same model, and the other one (MobiliSense) is filed under *Respiratory*. In the other direction every clinical row is empty of measurement-campaign work. The two literatures met on one page today mostly because a digest put them there.

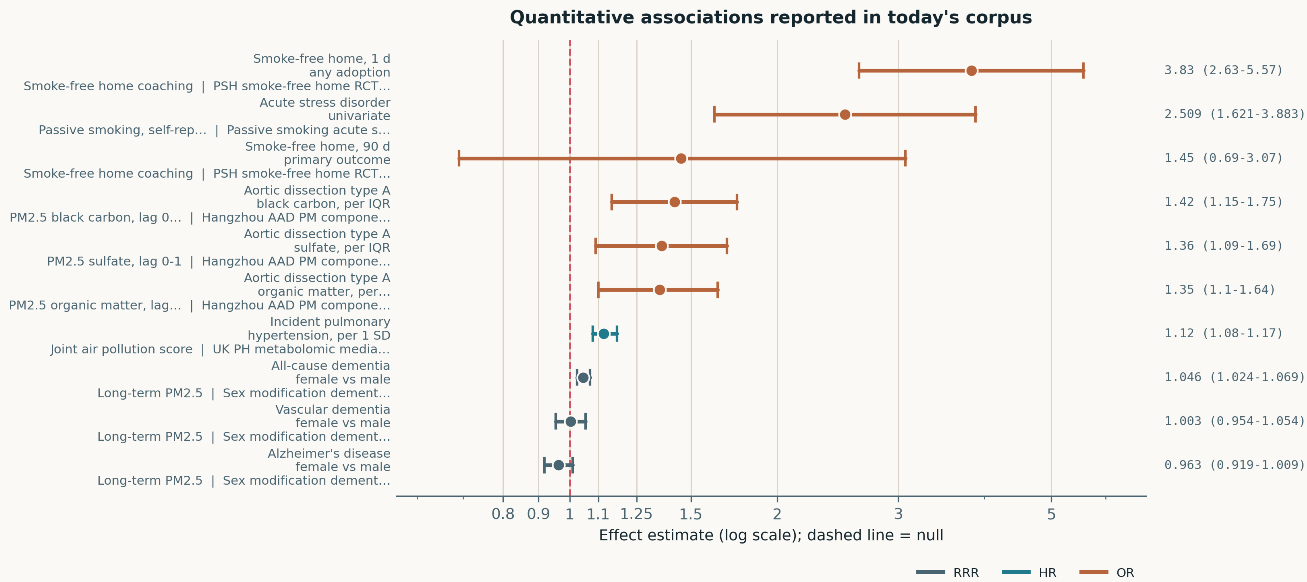


Fig. All ten quantitative estimates in the corpus on a common logarithmic axis. **The contrasts are not mutually comparable and three different quantities are plotted.** The three aortic-dissection ORs are per interquartile-range increments of distinct $PM_{2.5}$ components; the pulmonary-hypertension HR is per 1 SD of a composite pollution score; the three *RRR* points are *ratios of risk ratios* — female versus male — so for those the null line means **no sex difference**, not no effect, and reading them as small associations would be wrong. The two lowest-precision entries are interventions, not exposures: the smoke-free-home trial's primary outcome (OR 1.45, 0.69–3.07) crosses the null and its any-adoption secondary (3.83, 2.63–5.57) does not. The passive-smoking estimate is the univariate one, shown because the paper's own multivariate interval (1.959–3.144 around OR 1.990) is internally inconsistent. The trial's CO-verified abstinence OR of 6.94 (1.69–28.45) is omitted: it rests on 14 events total and would have compressed every other estimate on the axis.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Entries marked METADATA ONLY had no abstract in existence on the entry date; for these the entry states what the record establishes and what it does not, and asserts no finding.

Sensing, forecasting & instrumentation 7 records

Genoa absorption intercomparison 2026 Atmosphere

AE33 aethalometer against two filter-based absorption techniques in a coastal city

Multi-wavelength field intercomparison at the Multedo urban coastal site, Genoa, where the aerosol mixes traffic, shipping, industry and marine air. The AE33 was run against the Multi-Wavelength Absorption Analyzer and the Broadband Light Analyzer of Complex Aerosols across UV, visible and near-IR. Correlations are excellent — R^2 **0.93–0.99**, tightest between the two offline methods — but AE33 reports absorption coefficients **12–22% higher**, with the offset varying by wavelength. That pattern implicates the instrument's default multiple-scattering correction, which is tuned to a generic aerosol and not to this one. Absorption Ångström exponents from all three sit near unity, consistent with dominant primary combustion. The limitation is that one site and one campaign cannot fix the correction factor generally — which is precisely the point: a site-specific bias is not a constant you can publish once.

doi: 10.3390/atmos17090883

RI-URBANS urban AQ mapping 2026 Atmos Meas Tech

What a continent of low-cost-sensor and citizen-science air-quality pilots actually taught

Synthesis of the RI-URBANS mapping and hotspot-identification pilots across European cities, covering stationary low-cost sensor networks, mobile monitoring platforms and citizen-collected data alongside reference networks and modelling. The conclusion the programme reaches is organisational rather than technical: the limiting factor on producing comparable urban maps was **harmonisation between methods**, not the accuracy of any one method, and harmonisation was one of the project's stated primary aims for that reason. Two operational findings are worth carrying into any deployment — campaign design and sustained, direct communication with citizen participants determined data quality as much as hardware did. As a programme-level synthesis it reports no single effect estimate and cannot be appraised like a primary study; its value is the negative result about where the effort goes.

doi: 10.5194/amt-19-5729-2026

Alkylamine IC-MS/MS method 2026 J Chromatogr A

Speciated atmospheric alkylamines at picogram levels in gas and particle phase

Alkylamines drive new particle formation and early aerosol growth, and are analytically awkward — strongly polar, present at trace level, prone to adsorption and decomposition. This method pairs alkaline purging with acidic absorption for selective enrichment ahead of ion chromatography–tandem MS. Method detection limits are **9–94 $\mathrm{pg\,m^{-3}}$ gaseous and 0.1–1.1 $\mathrm{pg\,m^{-3}}$ in PM**, a substantial gain on existing offline methods, and the approach resolves the dimethylamine/monoethylamine isobaric pair and quantifies tertiary amines that derivatisation-based methods structurally cannot. Field-applied at septic tanks and traffic tunnels in Dalian, it recovered a broader speciation profile than previous work and seasonal variation in each species. It is an offline enrichment method, so time resolution is whatever the sampling interval is — this does not compete with online amine measurement for process studies, it competes on speciation and sensitivity.

doi: 10.1016/j.chroma.2026.467416

Baltic ship plume OFR aging 2026 Environ Sci Technol Lett

Fresh and photochemically aged emission factors from individual ships, measured at sea

An oxidation flow reactor operated from a moving research vessel, sampling identified ocean-going ships at **0.6–1.6 km** standoff in the Baltic, plus the research vessel's own plume as a control. Gas-phase and potential-aerosol-mass emission factors were measured fresh and after photochemical aging. **Particle mass rose after aging for every vessel sampled**, and secondary aerosol formation potential scaled with SO_2 emission — i.e. with fuel sulfur content, consistent with sulfate-driven particle formation. This is a pilot with a small vessel sample and no size-resolved chemistry reported in the abstract, so the emission factors are demonstrative rather than inventory-grade. The contribution is the framework: stack-level aging experiments and fixed ambient monitoring have both existed for years, and this joins them on a platform that can follow an individual, identifiable source.

doi: 10.1021/acs.estlett.6c00695

India winter PM2.5 seasonal forecast 2026 Sci Adv

Winter $PM_{2.5}$ severity over northern India is predictable a season ahead

Over a decade of quality-controlled Indian government reference $PM_{2.5}$ monitoring, used to establish that **roughly 70% of interannual variance** in northern Indian winter $PM_{2.5}$ is explained by western disturbances — Mediterranean-origin extratropical storms — acting through ventilation and wet removal. The storms in turn track to the position and strength of the subtropical jet, which is traceable to North Atlantic and tropical Indian Ocean sea-surface-temperature anomalies in the preceding autumn. Using only preseason SST, the authors predict winter WD precipitation and $PM_{2.5}$ severity one season ahead at r^2 **0.52–0.69**. The honest caveat is that this is meteorological predictability, not emissions predictability: it forecasts how well a given emission load will be ventilated, and a decade of records is a short sample for an interannual teleconnection. For network planning it is nonetheless actionable — it says which winters to instrument heavily.

doi: 10.1126/sciadv.aee4549

US precipitation washout DoubleML 2026 Research Square

Observation-derived washout coefficients, and how wildfire smoke breaks them

Preprint. Ten years of hourly EPA observations — 760 $PM_{2.5}$ and 451 PM_{10} monitors across the contiguous US — merged with NLDAS-2 meteorology and NOAA HMS smoke classifications, with a double machine-learning causal framework separating precipitation from confounding meteorology. Removal rises with rainfall intensity, particle size and initial concentration, and **falls as wildfire smoke loading rises** — smoke aerosol is less efficiently scavenged, which matters more each year. The authors fit power-law washout coefficients ($a = 0.0404$, $b = 0.5951$ for $PM_{2.5}$; $a = 0.0812$, $b = 0.4508$ for PM_{10}) intended to drop straight into chemical transport models. The inference threat is the usual one for DoubleML on observational data: unmeasured confounding is assumed away rather than eliminated, and hourly monitor-scale precipitation from a reanalysis is a coarse proxy for what fell on the sampler.

doi: 10.21203/rs.3.rs-10855101/v1

Open-pit mine dust stacking model 2026 Atmos Pollut Res

METADATA ONLY — multi-point cold-season dust prediction at an open-pit coal mine

No abstract existed in Crossref queried per-DOI, in Europe PMC or in PubMed on the entry date. What the record establishes: a multi-point dust-concentration prediction task at an open-pit coal mine during the cold season has been addressed with a stacking ensemble and an interpretability analysis. What it does not establish: the number and siting of monitoring points, the prediction horizon, the base learners in the stack, the interpretability method, any error figure, or the mine's location — none of which is inferable from the title and none of which is asserted here. Flagged for retrieval; open-pit mining dust is one of the few settings where a dense low-cost network has an unambiguous operational customer.

doi: 10.1016/j.apr.2026.103196

Exposure assessment & modelling 4 records

Hong Kong GEMA PM2.5 mobility 2026 Environ Res

1.7 million personal $PM_{2.5}$ measurements against momentary stress and activity

Two-day geographic ecological momentary assessment on 800 Hong Kong adults, integrating **1.7 million personal $PM_{2.5}$ measurements**, 2.1 million accelerometer records, 4,800 momentary assessments and eye-level greenspace derived from 4.3 million street-view images, summarised over 5- to 600-minute windows and analysed across eight activity-travel contexts with a DAG-guided multi-relation double-ML framework. Higher $PM_{2.5}$ tracked more light-intensity and less moderate-to-vigorous activity overall, and predicted worse *perceived* air quality; perceived pollution in turn predicted higher stress while measured $PM_{2.5}$ did not. The result that should travel: **context-specific estimates ran in opposite directions** while cumulative associations with stress showed nothing sustained. Two days per participant is thin for a stress outcome and reverse causation between activity and exposure is unresolved, but the decomposition itself is the finding — a pooled null here is an artefact of aggregation.

doi: 10.1016/j.envres.2026.125659

NZ tractor fleet PM inventory 2026 Sci Total Environ

An uncertainty-propagated $PM_{2.5}$ inventory for an entirely unregulated source class

Non-road mobile machinery is under-regulated and rising in relative importance as on-road sources are controlled; New Zealand is an instructive case because it pairs ultra-low-sulphur diesel and tightening on-road standards with *no* NRMM emission regulation at all. The inventory joins row-level registration microdata for 43,367 tractors, a licensing table defining 31,253 active units, and 29,278 dealer sales records (2017–2025), under an IPCC Tier 2 fuel-anchored framework with EMEP/EEA factors, tier hours calibrated against a CARB monitoring study, and fuel anchored to a 204 Ml national consumption target. 10,000-iteration Monte Carlo gives **115 t $PM_{2.5}$ (95% UI 73–190)** and 2258 t NO_x. The distributional result is the useful one: **Gini 0.51 for $PM_{2.5}$, with the dirtiest 30% of tractors producing 69% of it**, and modernisation running at only +2.7 percentage points a year. Activity hours remain the weakest link — they are calibrated, not measured.

doi: 10.1016/j.scitotenv.2026.182302

Paris Saharan dust heatwave 2026 Atmos Chem Phys

Dust aerosol cools Paris by up to 1 °C — and that does not mean comfort

Three Meso-NH simulations of the 15–19 June 2022 Paris heatwave — no dust, CAMS reanalysis dust, and doubled dust — validated against the PANAME-Urban campaign, which the model reproduces for aerosol optical depth and incoming solar radiation, with improved air temperature and boundary-layer height once aerosol is included. The dust plume's radiative effect cuts surface shortwave by up to **75 W m^{-2}** and air temperature by up to **1 °C**. The result worth the page is the one that refuses the easy conclusion: thermal comfort improves by up to 1 °C in *sunlit* urban and suburban areas at 16:00 UTC, but in shaded suburban areas the increase in diffuse radiation and humidity plus the drop in wind speed cancels the temperature benefit entirely. A single five-day episode in one city, and comfort indices are not health outcomes — but the case for coupling aerosol forecasts to heat-health warning systems is made on the disaggregation, not on the mean.

doi: 10.5194/acp-26-12729-2026

Construction equipment PM-bound EPFR 2026 Atmos Environ

METADATA ONLY — persistent free radicals on PM from non-road construction plant

No abstract retrievable on the entry date from Crossref per-DOI, Europe PMC or PubMed. What the record establishes: PM-bound environmentally persistent free radicals and downstream reactive radical species have been measured from non-road construction equipment under *real-world* rather than dynamometer conditions, which is the harder and rarer measurement. What it does not establish: radical concentrations or lifetimes, the engine tiers and duty cycles sampled, the EPR or spin-trapping protocol, or the site — none is asserted here. Relevant because EPFR content is a redox-activity metric that mass concentration does not predict, and construction plant sits inside the urban domain that dense networks are deployed to characterise.

doi: 10.1016/j.atmosenv.2026.122353

Cardiovascular & metabolic 2 records

Hangzhou AAD PM components 2026 J Am Heart Assoc

Which $PM_{2.5}$ component triggers acute aortic dissection, and by how much

Time-stratified case-crossover on Hangzhou acute aortic dissections, 2015–2023, with conditional logistic regression for single components and weighted quantile sum for the mixture. At **lag 0–1 days**, interquartile-range increases in black carbon, organic matter and sulfate were associated with **42% (OR 1.42, 1.15–1.75)**, 35% (1.35, 1.10–1.64) and 36% (1.36, 1.09–1.69) higher odds of Stanford *type A* dissection; type B associations were positive but not significant. In the joint model **black carbon carried more than half the weight**. Age, smoking, alcohol use and hypertension modified the associations. The case-crossover design self-controls for time-invariant confounding, which is its strength; the weaknesses are that component concentrations are strongly collinear (WQS mitigates but does not resolve this), exposure is assigned from ambient monitors rather than personally, and a single city's aerosol composition determines which component appears to dominate.

doi: 10.1161/jaha.126.049278

UK PH metabolomic mediation 2026 Ecotoxicol Environ Saf

Air pollution, pulmonary hypertension, and how much of it runs through metabolism

Prospective cohort of **444,346** participants free of pulmonary hypertension at baseline with NMR metabolomics, 2,328 incident cases over a 13.58-year median follow-up. $PM_{2.5}$, PM_{10}, NO_2 and NO_x entered individually and as a joint air pollution score. The score gave **HR 1.12 per SD (95% CI 1.08–1.17)**; individual pollutants ran 1.06–1.93 per 10 $\mu g/m^3$. Elastic-net regression produced a 105-metabolite signature dominated by lipoprotein measures, fatty acids and amino acids, itself associated with incident PH (HRs 1.12–1.24 per SD), **mediating 11.74% (7.91–18.95%)** of the score–PH association. The mediation estimate is the part to hold loosely: metabolites were measured once at baseline, so the mediator is cross-sectional with respect to a 13-year outcome, and the causal ordering of exposure, metabolome and disease cannot be established from that design. Residential exposure models add the usual non-differential error, which biases the 11.74% downward.

doi: 10.1016/j.ecoenv.2026.120778

Neuro, cognition & mental health 2 records

Sex modification dementia meta 2026 Environ Pollut

126 million participants: sex does not modify the $PM_{2.5}$–Alzheimer's association

Systematic review and meta-analysis of 24 longitudinal cohorts, **over 126 million participants**, searched to September 2025, using the female-to-male ratio of risk ratios as the modification metric — the correct estimand, and one that many sex-difference papers substitute a pair of stratified estimates for. For Alzheimer's disease the pooled $PM_{2.5}$ **RRR is 0.963 (95% CI 0.919–1.009)** and for vascular dementia **1.003 (0.954–1.054)**: null, and precisely so. The one positive is all-cause dementia (1.046, 1.024–1.069), a 4.6% relative excess in women that the authors decline to build policy on. Heterogeneity was negligible to low except PM_{10} in Alzheimer's ($I^2 = 61\%$), no publication bias was detected, and GRADE certainty is moderate. The stated limitation is the one that matters: methodological heterogeneity in how the constituent cohorts defined and adjusted sex could mask a true difference, so this is evidence against current *claims*, not proof of no effect.

doi: 10.1016/j.envpol.2026.129138

Passive smoking acute stress 2026 Front Public Health

Self-reported secondhand smoke and acute stress in 4,460 rescue personnel

Cross-sectional survey of 4,460 grassroots rescue officers and soldiers, screening with the Acute Stress Disorder scale and the PCL-5. Univariate odds were **2.509 (1.621–3.883)** for acute stress disorder and 3.298 (1.459–7.451) for PTSD; on full adjustment the ASD association survived and **the PTSD association did not** (2.513, 0.900–5.151). Among never-smokers, exclusive passive exposure still carried OR 1.910 (1.071–3.408). Three problems limit what can be drawn from this. Exposure is self-reported with *no particle measurement of any kind*, so the dose is unknown and likely differentially misreported by people who are already distressed. The design cannot order exposure and outcome. And the reported multivariate ASD interval — 1.959–3.144 around a point estimate of 1.990 — is internally inconsistent, so the univariate estimate is the one plotted here. Filed as a signal that SHS exposure belongs in trauma screening, not as a dose-response result.

doi: 10.3389/fpubh.2026.1790527

Respiratory & allergic 2 records

MobiliSense ambulatory spirometry 2026 Research Square

Sub-hourly personal black carbon and a measurable drop in FEV_1

Preprint. 199 adults in metropolitan Paris, 2018–2020, each carrying two portable pollutant monitors, a GPS receiver and an accelerometer through ordinary daily mobility, with **2,504 spirometry tests** taken morning and evening over three days. Mixed-effects models tested lags from 15 minutes to 6 hours. In multipollutant models, a 1 $\mu g/m^3$ increase in BC one hour before testing was associated with FEV_1 lower by **0.016 L (95% CI -0.024 to -0.008)**, and at two hours by **0.021 L (-0.034 to -0.007)**; $PM_{2.5}$ at 15, 30 and 60 minutes reduced FEV_1/FVC by 0.60, 0.70 and 0.50 percentage points. O_3 ran positive, which is the expected artefact of O_3 anticorrelating with traffic. The threats are real: 199 participants, three days, no personal-monitor validation reported against reference instruments, and spirometry quality in the field is hard. But this is the design that resolves an exposure window a fixed monitor cannot see at all.

doi: 10.21203/rs.3.rs-10386483/v1

Cardiorespiratory review 2026 Med Gas Res

Narrative review of air pollution and cardiorespiratory disease

Orientation-level review of the mechanisms linking ambient air pollution to cardiometabolic risk and cardiovascular death, with sections on heart failure, COPD and asthma exacerbation, on vulnerable groups (pregnancy, neonates, children, pre-existing cardiorespiratory disease), on the COVID-19 lockdown natural experiment, on international divergence in air quality standards, and on mitigation from policy through technology. There is no systematic search protocol, no PROSPERO registration, no pooled estimate and no risk-of-bias assessment, so it cannot be cited as evidence for any magnitude — and it does not claim to be. Listed because the lockdown-period material and the standards comparison are a useful map of where the regulatory targets currently sit, and because naming it here prevents it being mistaken for a systematic review later.

doi: 10.4103/mgr.MEDGASRES-D-25-00201

Reproductive & developmental 2 records

Prenatal greenspace ASD window 2026 Environ Pollut

A mid-to-late pregnancy window for green space — and $PM_{2.5}$ is not the mechanism

1:3 age- and sex-matched case-control from a tertiary maternal and child health hospital in central China: 207 children with autism spectrum disorder, 621 controls. Monthly residential NDVI and $PM_{2.5}$ were estimated from 10 months before to 3 months after birth and entered a distributed lag nonlinear model, with product terms testing NDVI × $PM_{2.5}$ interaction, bootstrapped mediation, and Bonferroni correction. Cases had lower NDVI in *all 14 months*; the protective window localises to mid-to-late pregnancy, roughly 6 to 1 month before term birth. The negative result is the one that earns the entry: **$PM_{2.5}$ neither mediated nor modified the green-space effect** after correction, which closes the most obvious alternative explanation rather than leaving it hanging. Caveats are the standard ones for this design — residential address as the exposure proxy, hospital-based case ascertainment, 207 cases against a 14-month lag structure, and green space as a proxy for an unmeasured bundle (noise, activity, socioeconomic position).

doi: 10.1016/j.envpol.2026.129118

Preeclampsia nationwide 2018-23 2026 Eur J Intern Med

METADATA ONLY — nationwide air pollution and preeclampsia, 2018–2023

The PubMed record carries no abstract, and neither Crossref nor Europe PMC held one on the entry date. What the record establishes: a nationwide study covering 2018–2023 has examined the association between ambient air pollution and preeclampsia. What it does not establish: the country, the pollutant set, whether PM was resolved at all, the case count, the exposure-assignment method, the trimester windows examined, or the direction and magnitude of any association — none of which follows from the title, and none of which is asserted here. Flagged for retrieval. Hypertensive disorders of pregnancy are one of the better-powered PM endpoints and a nationwide six-year series would be a substantial addition if the exposure model holds up.

doi: 10.1016/j.ejim.2026.107175

Occupational & indoor exposure 3 records

PSH smoke-free home RCT 2026 JAMA Netw Open

A smoke-free-home trial that missed its primary outcome, reported plainly

Cluster-randomised trial across 40 permanent-supportive-housing sites in the San Francisco Bay Area, 400 residents (mean age 54.5, 63% male) who smoked at home, January 2022 to March 2025, intervention sites receiving one-to-one coaching on adopting a smoke-free home and staff at all sites receiving brief cessation training. **The primary outcome failed**: 90-day smoke-free-home adoption was 6.8% versus 4.8%, **OR 1.45 (0.69–3.07)**. What moved was the softer secondary — any adoption of at least one day, 63.4% versus 36.8%, **adjusted OR 3.83 (2.63–5.57)** — and CO-verified 7-day abstinence, 6.3% versus 1.0%, OR 6.94 (1.69–28.45), which rests on *14 events in total* and should not be read as an effect size. The honest reading is that brief coaching moves intention and not maintenance in a population with very high baseline exposure, and that a waiting-list control cannot rule out attention effects on the self-reported outcomes.

doi: 10.1001/jamanetworkopen.2026.32697

Dutch school air cleaner RCT 2026 Front Public Health

Protocol: 180 classrooms, HEPA versus ionisation versus nothing

Protocol only, no results (ClinicalTrials.gov NCT07479420). 180 classrooms in 29 Dutch primary schools, clustered within schools and randomised to HEPA-filter portable air cleaners, ionisation/plasma cleaners, or none. Units were laboratory pre-tested, safety screened and operated at comparable clean air delivery rates — which is the design choice that makes the comparison mean something, since CADR is the confounder that sinks most field air-cleaner studies. The schedule is a three-week baseline or two-week post-intervention control plus three repeated three-week intervention periods, to 14 weeks per school. Airborne dust is collected passively on electrostatic dustfall collectors and assayed for bacterial and viral markers, with active sampling in 12 classrooms to validate the passive method; PM_{10}, PM_4, $PM_{2.5}$, PM_1, CO_2, temperature, humidity and VOCs are logged continuously. Absenteeism and parent-reported symptoms are retrospective, which is the weakest element. Worth tracking to results — ionisation devices rarely get a fair, CADR-matched test.

doi: 10.3389/fpubh.2026.1892902

CFRP inhalation risk review 2026 Regul Toxicol Pharmacol

Carbon-fibre composite dust: fibre-like fragments, ultrafines, and thin evidence

Life-cycle review of inhalation risk from carbon fibre-reinforced plastics across manufacturing, machining, recycling and end-of-life processing. Mechanical processing generates respirable high-aspect-ratio fragments; thermal processing generates ultrafine particles together with hazardous gases. The review's central judgement is a refusal: available evidence **does not demonstrate a magnitude or pattern of inhalation toxicity comparable to asbestos or to pathogenic high-aspect-ratio carbon nanotubes**, but it is too limited and heterogeneous — particularly for chronic exposure — to support any quantitative relative-toxicity comparison in either direction. It identifies the actual gaps: real-world occupational exposure data, harmonised emission scenarios, internal dose and biopersistence. As CFRP volumes rise with wind-blade and aerospace retirement, this is a source class whose workplace aerosol nobody is routinely monitoring.

doi: 10.1016/j.yrtph.2026.106243

Burden, policy & mitigation 3 records

Flue-gas condensable PM removal 2026 J Hazard Mater

Removing the particulate mass that filterable-PM controls never see

Condensable particulate matter leaves coal-fired and municipal-waste-incineration stacks as vapour and forms particles in the atmosphere downstream, which is why filterable-PM control devices and, in many jurisdictions, the compliance measurement itself miss it entirely. This bench study passes flue gas through a string-of-beads liquid and reports **29.56–65.27% overall CPM removal at 1.17–8.25% of the energy** of conventional condensation. The organic fraction is removed more readily than the inorganic throughout, with the sulfate fraction best at **78.34%**. Mechanistically, condensation dominates at high flue-gas temperature and dissolution takes over as the gas cools, and a two-term model combining them reproduces the measured removal. It is bench scale with synthetic conditions — no long-duration fouling, no liquid regeneration economics, no full-scale pressure drop — so the efficiency range is an upper bound on what a retrofit would deliver.

doi: 10.1016/j.jhazmat.2026.143505

Turkiye stroke GBD 2000-2023 2026 Neuroepidemiology

Ambient PM is the leading environmental risk for stroke death in Türkiye

GBD 2023 secondary analysis for Türkiye. Age-standardised stroke incidence fell from 141.47 to 103.05 per 100,000 between 2000 and 2023, deaths from 132.61 to 57.03, and DALYs from 2696.30 to 1073.51 — a genuine and large decline. In 2023, metabolic risks accounted for 38.24% of stroke deaths (high blood pressure alone 21.29%) and smoking 12.20% among behavioural risks; **ambient particulate matter led the environmental risks at 11.93%**, placing it within a percentage point of tobacco. Macroeconomic regressions gave mixed and partly counterintuitive results — universal health coverage service capacity correlated positively with death rates — which is a familiar ecological artefact of ascertainment improving with health-system capacity. The deeper limitation for this watch: GBD's PM exposure surface is itself a model, so the 11.93% is an attribution built on modelled exposure, not an independent measurement.

doi: 10.1159/ned/adwag030

Intergenerational mortality equity 2026 Environ Int

METADATA ONLY — intergenerational equity in acute and chronic mortality impacts

No abstract retrievable on the entry date from Crossref per-DOI, Europe PMC or PubMed. What the record establishes: mortality impacts of *both* acute and chronic air pollution exposure have been assessed under an intergenerational health-equity framing — an unusual decomposition, since acute and chronic burden estimates are normally produced by separate literatures with incompatible baselines. What it does not establish: the populations, the pollutant set, whether PM is resolved, the discounting or life-table assumptions that an intergenerational comparison entirely depends on, the geography, or any result. None is asserted here. Flagged for retrieval as the highest-value of the four metadata-only records in this issue.

doi: 10.1016/j.envint.2026.110515

Mechanistic toxicology 1 record

Tire wear particle phytotoxicity 2026 Curr Biol

Tire wear particles stunt a roadside grass and raise root oxidative stress

Controlled tire-wear-particle exposure of *Bromus catharticus*, a common roadside weed in Japan, testing an endpoint the TWP literature has largely skipped — previous work concentrated on aquatic systems and on crops, not on the roadside and urban-park vegetation that sits metres from the source. Both **shoot and root growth were significantly inhibited**, and root malondialdehyde — a lipid-peroxidation marker — rose significantly, indicating oxidative stress as at least one operative mechanism. The abstract does not give the applied dose, the particle size distribution, the exposure route (soil versus foliar) or the exposure duration, and all four determine whether the treatment is environmentally realistic; a single annual grass in a pot cannot stand for roadside vegetation generally. Carried here because TWP is the fastest-growing non-exhaust PM source, is poorly sized by optical monitors, and now has a documented terrestrial effect at the point of emission.

doi: 10.1016/j.cub.2026.06.087

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Passive smoking acute stress	Is passive smoking associated with acute stress response? An observational real-world study	*Front Public Health*	Cross-sectional survey	Self-reported secondhand smoke exposure, no measurement	China
Sex modification dementia meta	Does Reported Sex Modifies the Association between Ambient Air Pollution exposure and Alzheimer's Disease and Related Dementia? A Systematic Review and Meta-Analysis	*Environ Pollut*	Systematic review + random-effects meta-analysis	Long-term $PM_{2.5}$ and 11 further pollutants, F:M ratio	Global
Cardiovascular & metabolic					
Hangzhou AAD PM components	Associations of Fine Particulate Matter Components With Acute Aortic Dissection: A Time-Stratified Case-Crossover Study	*J Am Heart Assoc*	Time-stratified case-crossover	Component-resolved $PM_{2.5}$: BC, OM, sulfate, nitrate	China
UK PH metabolomic mediation	Long-term ambient air pollution exposure, metabolomic signatures, and incident pulmonary hypertension: A large prospective cohort study	*Ecotoxicol Environ Saf*	Prospective cohort (mixtures) + omics mediation	$PM_{2.5}$, PM_{10}, NO2, NOx and a joint air pollution score	UK
Reproductive & developmental					
Preeclampsia nationwide 2018-23	Association between air pollution and preeclampsia: a nationwide study from 2018 to 2023	*Eur J Intern Med*	Metadata only (no abstract)	Ambient air pollution, nationwide registry linkage	Not stated
Prenatal greenspace ASD window	Mid-to-Late Pregnancy: A Critical Window for the Protective Effect of Green Space Against Autism Spectrum Disorder	*Environ Pollut*	Matched case-control + DLNM	Monthly residential $PM_{2.5}$ alongside NDVI, -10 to +3 months	China
Respiratory & allergic					
Cardiorespiratory review	Breathing in harm: current insights into the effects of air pollution on cardiorespiratory health	*Med Gas Res*	Narrative review	Ambient air pollution, no single exposure metric	Global
MobiliSense ambulatory spirometry	Evaluation of the effects of air pollutants on lung function using ambulatory monitor data from the MobiliSense Project	*Research Square*	Personal exposure monitoring + panel spirometry	Continuous personal BC and $PM_{2.5}$ with GPS, 15 min-6 h lags	France
Mechanistic toxicology					
Tire wear particle phytotoxicity	Tire wear particles impact a roadside wild plant species	*Curr Biol*	Controlled exposure (ecotox)	Tire wear particles, non-exhaust traffic source	Japan
Occupational & indoor					
CFRP inhalation risk review	Inhalation risks of carbon fiber-reinforced plastic-derived emissions: Occupational exposure and regulatory perspectives	*Regul Toxicol Pharmacol*	Critical review	Respirable CFRP fibre fragments and thermal ultrafines	Global
Dutch school air cleaner RCT	Evaluating air cleaner effectiveness in schools: a comprehensive review and protocol of a cluster-randomized controlled trial of physicochemical and microbial markers	*Front Public Health*	Cluster-randomised trial protocol	Continuous PM_{10}, PM4, $PM_{2.5}$, PM1 plus CO2, T, RH, VOC	Netherlands
PSH smoke-free home RCT	Smoke-Free Home Intervention in Permanent Supportive Housing: A Cluster Randomized Clinical Trial	*JAMA Netw Open*	Cluster-randomised controlled trial	Secondhand smoke in multiunit housing, CO-verified	USA
Sensing, forecasting & instrumentation					
Alkylamine IC-MS/MS method	An accurate and sensitive method for identifying atmospheric alkylamines: selective enrichment by alkaline purging-acidic absorption coupled with ion chromatography-tandem mass spectrometry	*J Chromatogr A*	Analytical method validation	Speciated gas and PM-phase alkylamines by IC-MS/MS	China
Baltic ship plume OFR aging	In Situ Collection and Photochemical Aging of Individual Ship Plumes Captured at Sea	*Environ Sci Technol Lett*	Field campaign / shipborne	Shipborne oxidation flow reactor on individual vessel plumes	Europe
Genoa absorption intercomparison	Multi-Wavelength Optical Characterization of Carbonaceous Aerosols at a Coastal Urban Site in Genoa: Intercomparison of Real-Time and Filter-Based Techniques	*Atmosphere*	Multi-instrument field evaluation	AE33 aethalometer vs MWAA and BLAnCA filter analysis	Italy
India winter $PM_{2.5}$ seasonal forecast	Seasonal predictability of winter $PM_{2.5}$ pollution severity in India through a strong pollution-extratropical storm connection	*Sci Adv*	Seasonal prediction framework	Government reference $PM_{2.5}$ network, >10 winters	India

First author	Focus	Journal	Design	Metric	Region
Open-pit mine dust stacking model	Multi-Point Prediction and Interpretable Analysis of Dust Concentration in Open-Pit Coal Mine During the Cold Season Using a Stacking Learning Framework	*Atmos Pollut Res*	Metadata only (no abstract)	Multi-point cold-season mine dust concentration	Not stated
RI-URBANS urban AQ mapping	Enhancing Urban Air Quality Mapping through Novel Measurement and Modelling approaches, and Citizen Science: Actionable Insights from the RI-URBANS Project	*Atmos Meas Tech*	Multi-platform observation synthesis	Low-cost sensor networks, mobile platforms, citizen data	Europe
US precipitation washout DoubleML	Particulate matter removal by precipitation under normal and wildfire conditions across the United States using double machine learning	*Research Square*	Causal ML on monitoring records	Hourly EPA $PM_{2.5}$ and PM_{10} vs NLDAS-2 precipitation	USA
Exposure assessment & modelling					
Construction equipment PM-bound EPFR	Real-world emissions of PM-bound environmentally persistent free radicals and reactive radical species generation from non-road construction equipment	*Atmos Environ*	Metadata only (no abstract)	PM-bound environmentally persistent free radicals	Not stated
Hong Kong GEMA $PM_{2.5}$ mobility	Spatiotemporal associations between $PM_{2.5}$, greenspace, physical activity, and stress: a geographic ecological momentary assessment study in Hong Kong	*Environ Res*	Geographic ecological momentary assessment	Mobile personal $PM_{2.5}$, 5-600 min cumulative windows	China
NZ tractor fleet PM inventory	A calibrated, uncertainty-aware, and spatially explicit emission inventory of New Zealand's agricultural diesel tractor fleet	*Sci Total Environ*	Emission inventory	Non-road machinery $PM_{2.5}$, Monte Carlo uncertainty	New Zealand
Paris Saharan dust heatwave	Numerical study of a Saharan dust plume impact on radiation, urban climate and thermal comfort in the Paris region during an heatwave event	*Atmos Chem Phys*	Mesoscale simulation + campaign validation	CAMS dust aerosol in Meso-NH, AOD-validated	France
Burden, policy & mitigation					
Flue-gas condensable PM removal	Study on the removal of condensable particulate matter from flue gas using string-of-beads liquid	*J Hazard Mater*	Bench / device experiment	Condensable PM from coal and MSW incineration flue gas	China
Intergenerational mortality equity	Assessing intergenerational health equity in the mortality impacts of acute and chronic air pollution exposure	*Environ Int*	Metadata only (no abstract)	Acute and chronic exposure, mortality impact attribution	Not stated
Turkiye stroke GBD 2000-2023	Trends and Determinants of Stroke Burden in Turkiye, 2000-2023: An Analysis Using Global Burden of Disease Data	*Neuroepidemiology*	GBD secondary analysis	GBD ambient particulate matter risk-attributable fraction	Turkiye

Provenance and method

Window. Entry date 9 September 2026 only. The 9 September scheduled run did not fire; this issue was built on 10 September at 18:39 local, which is before the 22:00 gate, so 9 September is the newest buildable date and 10 September is deliberately left unopened rather than harvested against a partial day.

Sources. PubMed E-utilities on `[Date - Entry]`, two axes run as separate queries (health 11 hits, sensing 1); the **PubMed connector** run independently on the same two axes returned 16 on the health axis, **nine PMIDs the E-utilities leg never returned**, three of which ship — the first run in which the connector contributed net-new records rather than confirming the harvester. **Europe PMC** `CREATION_DATE` (24 hits, including two preprints that ship). **Crossref by ISSN** across the watch's journal list (68 hits) — the only leg that reaches *Atmos. Meas. Tech.*, *Atmos. Chem. Phys.* and *Atmosphere*, which PubMed does not index and which supplied four of the seven sensing records. **arXiv** relevance query with a local recency screen — one record passed it (wildfire smoke and Californian crop yields) and proved to be already in the archive by title. **OpenAlex** returned HTTP 429 and contributed nothing; that leg has been degraded since July. **Consensus** returned three sensing hits: one is already in `seen.json` from 5 August, and the other two were checked against their Crossref *created* dates — 26 March and 20 January 2026 — so both sit far outside this window and are archive gaps rather than records for this issue.

Screening. 105 raw records, 102 after cross-source deduplication on PMID → DOI → normalised title, 14 already in `seen.json`, 88 fresh, plus 9 connector-only records. **26 in scope, 71 excluded**, each with a logged reason in `state/rejected.jsonl`. Nineteen exclusions are *Meas. Sci. Technol.* bycatch from the ISSN leg (bearing diagnostics, SLAM, inertial navigation) and eleven are non-atmospheric *ES&T*.

Metadata-only records. Four Elsevier records had no abstract in existence anywhere on the entry date — checked per-DOI against Crossref, against Europe PMC, and against PubMed `efetch`. Crossref indexes a DOI before the publisher deposits the abstract, and a one-day window sits inside that gap; this is the second consecutive issue to hit it and it should now be treated as expected behaviour for same-day Elsevier deposits, not as a fault. These four are unambiguously in scope on the title and ship as METADATA ONLY, tier C, asserting no finding. Two further abstract-free records were *rejected* because without an abstract their PM scope could not be confirmed at all.

Trial watch. ClinicalTrials.gov returned **0** PM or air-pollution registrations or updates on 9 September against **1,189 registry-wide** on the no-term control — a topical zero, not an outage. Accumulated trial watch surfaces in the weekly rollup (next: Saturday 12 September).

Labels and estimates. Subtopic, design, particle-metric and geography labels are assigned from title, abstract, keywords and MeSH; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Effect estimates are transcribed verbatim from abstracts and use heterogeneous contrasts — see the forest-plot caption, which also records the one estimate omitted and why. All DOIs resolve to the publisher of record and are checked automatically before build. Abstracts remain the copyright of their publishers.